■ RAINVISION INDIA

Regional Rainfall Prediction & Local Weather Outlook Web App

Open the live web app:
https://rainvisionindiawebapp.vercel.app

RainVision India is a mobile-friendly college project prototype designed to help users view a location-based rainfall outlook for the next 24 hours. Users can search for a city, town or village, or provide latitude and longitude coordinates to view the local forecast.

1. Project Overview

Project name	RainVision India
Project type	Regional rainfall prediction and weather-forecasting web application
Main purpose	Show an easy-to-understand local rain outlook for the next 24 hours
Target locations	Cities, towns and villages in India
Platform	Responsive web app accessible through a mobile or desktop browser
Project status	Educational college-project prototype

2. Main Features

• **Location search:** Enter a city, town or village name and request its forecast.

• **Coordinate-based forecast:** Enter latitude and longitude for a more specific location.

• **Optional device location:** GPS access is requested only when the user taps “Use my location”.

• **24-hour outlook:** Displays forecast information for the next 24 hours in the selected location’s local time.

• **Rainfall chart:** Shows rainfall by hour; users can tap a chart bar to inspect an hour.

• **Forecast indicators:** Includes estimated rainfall total, hours with rain, peak rain probability and average temperature.

• **Hourly weather details:** The table includes time, rain, chance of rain, temperature, humidity and cloud information.

• **CSV export:** Users can download the hourly forecast data as a CSV file.

3. How the Application Works

Step 1. The user enters a place name or supplies latitude and longitude coordinates.

Step 2. The application obtains the forecast for the selected location from its weather-data service.

Step 3. The web page presents the next 24 hours of forecast information, including rainfall and related weather indicators.

Step 4. The user can inspect hourly rainfall using the chart and download the forecast table as a CSV file.

4. Data Source and Prediction Approach

The live web page identifies Open-Meteo as its weather-data provider. The displayed forecast is based on a regional weather-model grid. The website presents this weather-service forecast as the main prediction.

The project also includes an optional machine-learning component in the original Python application: a **Random Forest model trained on historical data**. The training workflow remains in the Python version. The lightweight browser application does not claim to train a machine-learning model locally.

5. Technology and Components

Component	Role in the project
Web interface	Mobile-friendly browser interface for entering a location and viewing results
Weather-data service	Open-Meteo is identified by the website as the forecast data provider
Forecast visualisation	Hourly rainfall chart and 24-hour forecast table
Data export	CSV download for the hourly forecast
Python machine-learning version	Optional Random Forest training workflow using historical data
Deployment	Hosted as a web application on Vercel

6. Expected Uses

The prototype can help students and general users explore local hourly weather information, understand rainfall probability and estimated rainfall, and compare forecast conditions across locations. It can also serve as a demonstration of how weather data, visualisation and a separate machine-learning workflow can be combined in a regional rainfall project.

7. Limitations and Safety Note

This is an educational prototype, not an official weather-warning system. The forecast uses a regional weather-model grid and may miss highly localised showers. Forecast values can change as weather data updates. For safety-related decisions, users should check official India Meteorological Department (IMD) alerts.

8. Live Project Link

Click the link below to open RainVision India:
https://rainvisionindiawebapp.vercel.app

Prepared from the features and project notes visible on the live RainVision India website. This document describes the current web prototype and distinguishes it from the optional Python Random Forest workflow.